HL7 and XML

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HL7
Health Level 7

Organized to create standards for the exchange, management and integration of data that supports clinical patient care and the management, delivery and evaluation of healthcare services

- Founded by healthcare providers in 1987
- Version 1.0 late in 1987
- Version 2.0 late in 1988
- Versions 2.1, 2.2 and 2.3 published in 1990, 1994 and 1997; ANSI standards
- Pragmatic approach
- Work on Version 3 (XML-based) is ongoing

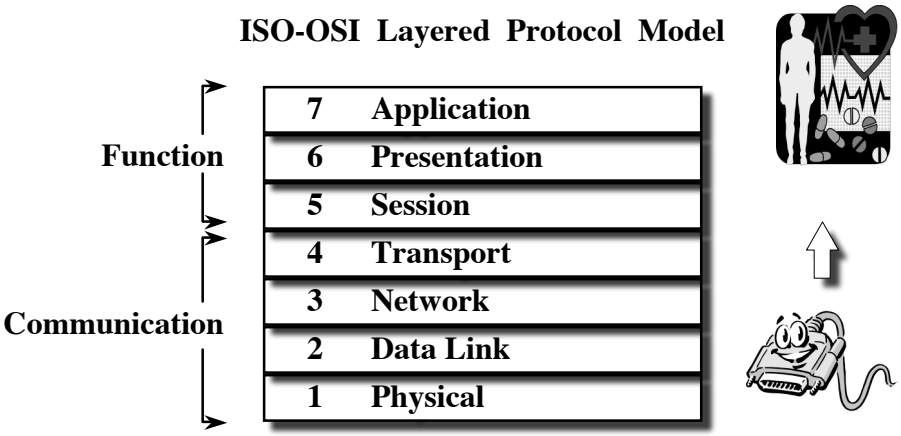
HL7 Standard

- Message-based protocol for exchange of healthcare information
- Defines message identity and general message format (fields)
- Published catalog of messages with identified fields

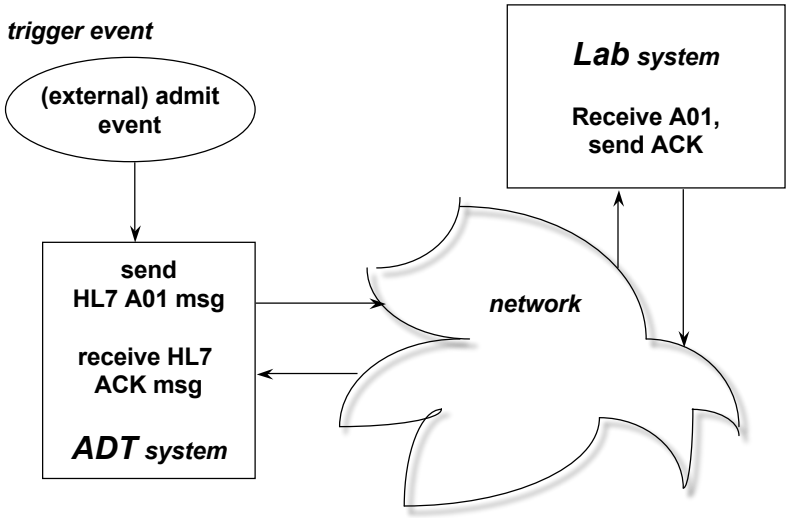
"Level Seven"

A protocol for the exchange of health care information

ISO-OSI Layered Protocol Model



HL7 Transactional Model



HL7 Abstract Messages

- Identifies data fields
- Specifies transmission success responses
- Describes transmission error conditions and responses
- DOES NOT describe the byte string contained in the message.

Admit Message

segments, fields, components & subcomponents

```
MSH|^~\&|ADT1|MCM|LABADT|MCM|198808181126|SECURITY|ADT^A01|MSG00001|P|2.3|<cr>
EVN|A01|198808181123||<cr>
PID|||PATID1234^5^M11||JONES^WILLIAM^A^III||19610615|M||C|1200 N ELM
   STREET^^GREENSBORO^NC^27401-1020|GL|(919)379-1212|(919)271-3434||S||
   PATID12345001^2^M10|123456789|987654^NC|<cr>
NK1|JONES^BARBARA^K|WIFE||||||NK^NEXT OF KIN<cr>
PV1|1|I|2000^2012^01||||004777^LEBAUER^SIDNEY^J.|||SUR||||ADM|A0|<cr>
```

Abstract Message Structure

Segments

[...] is 0 to 1, {...} is 1 to many, [{...}] is 0 to many

MSH	**Message header**
[{NTE}]	**Notes and comments**
[PID	**Patient identification**
[{NTE}]	**Notes and comments about the patient**
[{AL1}]	**Allergy data**
[PV1]	**Patient Visit**
{ ORC	**Common Order**
[Order Detail	**chosen from OBR, RXO, RQD [RQ1], {ODS}, {ODT}**
[{NTE}]	**Notes and comments about the order**
[{ OBX	**Observational results**
[{NTE}] }]	**Notes and comments about results**
]	
[BLG]	**Billing**
}	

Result Message

MSH, EVN, PID, PV1, ORC,
OBR||8974-9^BP Battery^LN|
OBX|1|CE|8357-6^METHOD^LN|M^Manual|
OBX|2|CE|8358-4^DEVICE^LN|1|AC^Adult Cuff|
OBX|3|CE|8359-2^SITE^LN|1|RBA^Rt Brachial Artery|
OBX|4|CE|8361-7^POSITION^LN|1|SIT^Sitting|
OBX|5|NM|8479-8^SBP^LN|1|138|mmHg|
OBX|6|NM|8462-4^DBP^LN|1|85|mmHg|

HL7 v2.x is not Plug and Play

- Cost of installing an HL7 interface: 2-4 weeks of analyst time

- Issues
  - > Different implicit information models
  - > Misunderstanding of specifications
  - > No vocabulary to describe conformance except by detailed specs
  - > Significant local demands on vendors

Variability in HL7 Interfaces

- Site 1:

  OBX|1|CE|ABO^**ABO GROUP**||O^**Type O**|
- Site 2:

  OBX|1|CE|BLDTYP^**ABO GROUP**||TYPEO^**Type O**|
- Site 3:

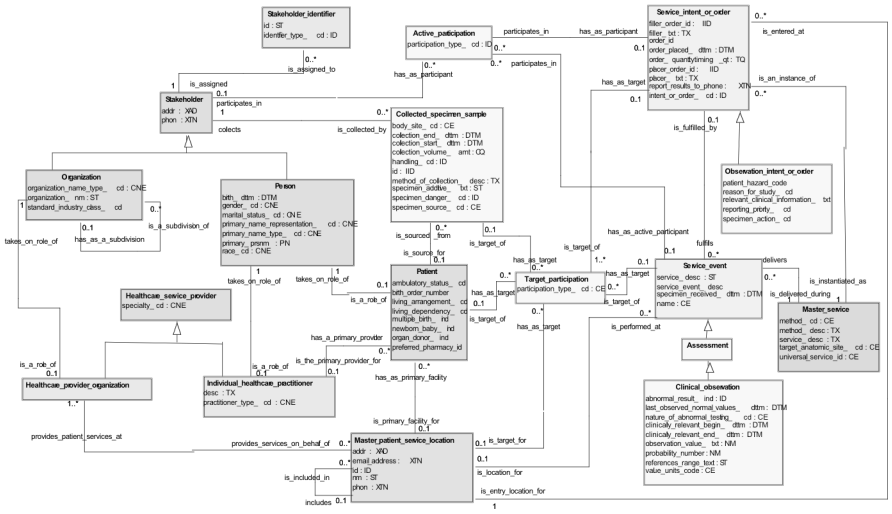
  OBX|1|CE|ABOTYPE^**ABO GROUP**||OPOS^**Type O**|

"when you've seen one HL7 interface you've seen one HL7 interface"

Goals for Version 3

- Substantially reduce interface development time
  - > Clarify spec for messages
  - > Specified information model
- Method for conformance specification
- Support modern communications infrastructures
- Reference Information Model (RIM)
  - > Coherent shared information model
  - > Includes all content of HL7 messages
  - > Provides consistency to messages across usage settings
  - > 120 defined classes (May '99)

Reference Information Model (RIM)



HL7 v3 Message Specification

How to get from the RIM to a specific message structure

- Message Information Model (MIM)
  - > Subset of the RIM contained in a specific message
- Heirarchical Message Description (HMD)
  - > "Recipes" for messages; define data relationships and message format
- Messages may be encoded in any of a number of schemes
- Encoding formats are described in Implementation Technology Specifications (ITS, XML is one)
- The information content of the message is identical regardless of the ITS

Advantages of XML for Message Formatting

- The syntax handles recursion and nesting
  - > Variably nested structures to arbitrary depth
  - > More flexible than segments, fields, components & subcomponents
- Objects (including contained objects) can be represented
- Relational structures can be represented
- Data files are nearly self-documenting (human readable)
- Software tools (parsers, etc.) are generally available
- A developing standard in data management and business communication

HL7 2.3 Message Format

```
MSH|^~\&|LABGL1||DMCRES||199812300100||ORU^R01|LABGL1199510221838581|P|2.3
     |||NE|NE
PID|||6910828^Y^C8||Newman^Alfred^E||19720812|M||W|25 Centscheap Ave^^
     Whatmeworry^UT^85201^^P||(555)777-6666|(444)677-7777||M||773789090
OBR||110801^LABGL|387209373^DMCRES|18768-2^CELL COUNTS+DIFFERENTIAL TESTS
     (COMPOSITE)^LN|||199812292128||35^ML||||||IN2973^Schadow^Gunther^^^^M
     D^UPIN
     ||||||||||^Once|||||CA20837^Spinosa^John^^^^MD^UPIN
OBX||NM|4544-3^HEMATOCRIT (AUTOMATED)^LN||45||39-49
     ||||F|||199812292128||CA20837
OBX||NM|789-8^ERYTHROCYTES COUNT (AUTOMATED)^LN||4.94|10*12/mm3
     |4.30-5.90||||F|||199812292128||CA20837
```

Figure 7-4. OBR attributes

```
MSH
PID
{ [ ORC ]
    { OBR
      { OBX }
    }
}
```

SEQ	LEN	DT	OPT	RP/#	TBL#	ITEM #	ELEMENT NAME
1	4	SI	C			00237	Set ID - OBR
2	22	EI	C			00216	Placer Order Number
3	22	EI	C			00217	Filler Order Number +
4	200	CE	R			00238	Universal Service ID
5	2	ID	X			00239	Priority

HL7 v3 Message Format


```
<Labrs3P00 T="Labrs3P00">
    <Labrs3P00.PTP T="PTP">
        <PTP.primrPrsnm T="PN">
            <fmn T="ST">Sample</fmn>
            <gvn T="ST">George</gvn>
            <mdn T="ST">H</mdn>
        </PTP.primrPrsnm>
    </Labrs3P00.PTP>
    <Labrs3P00.SIOO_L T="SIOO_L">
        <SIOO_L.item T="SIOO">
            <SIOO.filrOrdId T="IID">LABGL110801</SIOO.filrOrdId>
            <SIOO.placrOrdId T="IID">DMCRES387209373</SIOO.placrOrdId>
            <SIOO.InsncOf T="MSRV">
                <MSRV.unvSvcId T="CE">18768-2</MSRV.unvSvcId>
                <MSRV.svcDesc T="TX">CELL COUNTS+DIFFERENTIAL TESTS (COMPOSITE)</MSRV.svcDesc>
            </SIOO.InsncOf>
            <SIOO.SRVE_L T="SRVE_L">
                <SRVE_L.item T="SRVE">
                    <SRVE.name T="CE">4544-3</SRVE.name>
                    <SRVE.svcEvntDesc T="ST">HEMATOCRIT (AUTOMATED)</SRVE.svcEvntDesc>
                    <SRVE.CLOB T="CLOB">
                        <CLOB.obsvnValu T="NM">45</CLOB.obsvnValu>
                        <CLOB.refsRng T="ST">39-49</CLOB.refsRng>
                        <CLOB.clnRlvnBgnDtm T="DTM">199812292128</CLOB.clnRlvnBgnDtm>
                    </SRVE.CLOB>
                    <SRVE.spcmRcvdDtm T="DTM">199812292315</SRVE.spcmRcvdDtm>
                </SRVE_L.item>
        </SIOO_L.item>
    </Labrs3P00.SIOO_L>
</Labrs3P00>
```


XML Format for HL7 2.3.1 Messages

- An XML syntax and transformation scheme for current HL7 messages
  - > "This proposal ... addresses the process for translating HL7 Version 2.3.x message instances into XML documents that are valid with respect to the proposed XML DTD."
  - > "The ability to explicitly represent an HL7 requirement in XML confers the ability for message receivers to validate that requirement with an off-the-shelf XML parser."
- Interim strategy until vendors fully support v3
- Recognizes enterprise transition period to XML messaging

HL7 Clinical Document Architecture (CDA)

- Goal: a standard markup framework clinical documents
- Key issues
  - > Longevity of data
    - Applications must evolve but data must persist
    - Applications depreciate in value but data appreciates in value
    - Information design should outlive system design
  - > XML as a persistent data format that can move between systems: sharing data, integrating knowledge
  - > Mix narrative and data that naturally belong together
  - > Patient care documents are the priority in the standard
  - > Initial CDA designed for document exchange
- Version 1 released in 2000, version 2 balloted in 2004

Clinical Document Characteristics

- Persistence
  - > Defined by local and regulatory requirements
- Stewardship
  - > Maintained by an organization or person
- Authentication
  - > A collection of information that is to be legally authenticated
- Context
  - > Circumstances of creation and use
- Wholeness
  - > Legal authentication applies to the document as a whole and not to parts of the document out of context. The document also establishes a context for use of the contained information (the data in the document "belong together").
- Human readability, can be multimedia

Documents vs. Messages

	Documents	**Messages**
Lifetime	persistent	temporary
Communication	human-to-human	system-to-system
Relation to caregivers	care-givers are trained to create documents...	... but not messages
Legal aspects	have legal standing	signed? legally accepted?
Source	defined by precedent	designed per use case
Context	document as a whole	must be defined in each segment

CDA Markup Levels

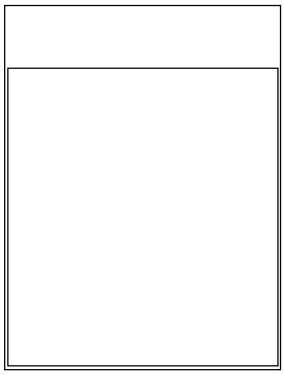
A multilevel representation of medical documents that can be passed as messages and which make up the medical record.

- Level 1: Unconstrained CDA schema
  - > CDA header; body may be plain or marked up
- Level 2: Constrained section markup
  - > Specific arrangement of sections defined by document type
- Level 3: Contrained “entries” or fields
  - > Tagging in text based on HL7 v.3 RIM
  - > Local tags in own namespaces

CDA General Structure

```
<ClinicalDocument>
    ... CDA Header ...
    <StructuredBody>
        <section>
            <text>...</text>
            <Observation>...</Observation>
            <Observation>
                <reference>
                    <ExternalObservation>
                    ...
                    </ExternalObservation>
                    </reference>
            </Observation>
        </section>
        <section>
            <section>...</section>
        </section>
    </StructuredBody>
</ClinicalDocument>
```

CDA Document with Unstructured Text



Header & "wrapper"

- –Document information
- –Encounter data
- –Service actors (such as providers)
- –Service targets (such as patients)

Body:
Clinical Document
as text

CDA Header Example

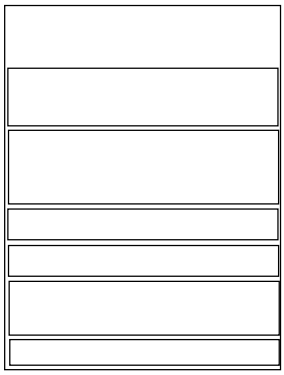
C:\KEG\CDA\CDA_Framework.final\CDA_minimal_samplenoDTD.xml - Microsoft Internet Explorer

File Edit View Favorites Tools Help Links

```
  <?xml version="1.0" ?>
  <!-- DOCTYPE levelone PUBLIC "-//HL7//DTD CDA Level One 1.0//EN" "levelone_1.0.dtd"  -->
- <levelone>
  - <clinical_document_header>
      <id EX="a123" RT="2.16.840.1.113883.3.933" />
      <document_type_cd V="11488-4" S="2.16.840.1.113883.6.1" DN="Consultation note" />
      <origination_dttm V="2000-04-07" />
    - <patient_encounter>
        <id EX="KPENC1332" RT="2.16.840.1.113883.3.933" />
        <practice_setting_cd V="GIM" S="2.16.840.1.113883.5.10588" DN="General internal
          medicine clinic" />
        <encounter_tmr V="2000-04-07" />
      </patient_encounter>
    - <legal_authenticator>
        <legal_authenticator.type_cd V="SPV" />
        <participation_tmr V="2000-04-08" />
        <signature_cd V="S" />
      + <person>
      </legal_authenticator>
    + <originator>
    + <originating_organization>
    + <provider>
    - <patient>
        <patient.type_cd V="PATSBJ" />
      - <person>
          <id EX="12345" RT="2.16.840.1.113883.3.933" />
        + <person_name>
        </person>
        <birth_dttm V="1932-09-24" />
        <administrative_gender_cd V="M" S="2.16.840.1.113883.5.1" />
      </patient>
    </clinical_document_header>
  - <body>
    - <section>
        <caption>History of Present Illness</caption>
```

My Computer

CDA with Section-Level Templates

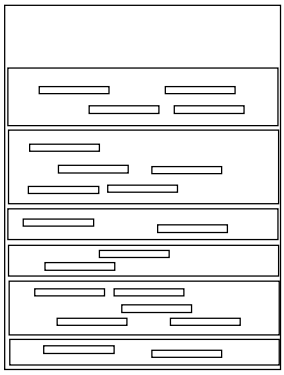


Header & "wrapper"

Body:
Clinical Document
with constrained sections

- Constraint based on document type

CDA with “Entry Level” Templates



Header & "wrapper"

Body:
Clinical Document
with constrained “entries”
(fields)

- Based on RIM
- Local elements may be added in their own namespaces

Key Header Elements

- ID, set ID, version, addendum vs. replacement
- Fulfills order
- Document type (LOINC for clinical observations)
- Origination time
- Confidentiality level
- Patient encounter
- Service actors (care providers; individuals and organizations)
  - > Authenticator, legal authenticator, originator, intended recipient, originating organization, provider, transcriptionist
- Service target (living or inanimate)
  - > If patient, one and only one

Structural Markup

- StructuredBody, component and section tags
- HTML-like (captions/headings, paragraphs, lists, tables)
- Recursive relationships
- Content tag: generic identifier and marker for text sequences
- Coded entry: standard vocabulary entry, can be targeted to a text span defined by content tags
- Observation and value tags
- Data elements and types from the RIM

References

- HL7 Organization
  > http://www.hl7.org
- HL7 XML Technical Committee
  > http://www.hl7.org/special/committees/sgml/sgml.htm
- Patient Record Architecture (PRA) tutorial
  > http://www.hl7.org/library/committees/sgml/PRATutorialHomePage.htm
- Dolin, Alschuler, et al. The HL7 Clinical Document Architecture. JAMIA 8:552-569, 2001.